COMP7503 Written Assignment 1: Temporal and Frequency Masking

Name: Su Chuanzhou | **University No.:** 3036837980 | **Course:** COMP7503 Multimedia Technologies

Method

The two clips were generated digitally rather than captured with a microphone; this was confirmed acceptable on the course forum (21 Oct 2026), provided the generation settings are stated, and they are given below. The clips were generated in Python using NumPy and SciPy. Sine tones were placed on a silent timeline, and overlapping tones were added together. Each tone had a 3 ms fade at its beginning and end to reduce clicks. The clips were exported as mono, 16-bit PCM WAV files at 44,100 Hz, without noise reduction, equalisation or compression. The waveforms and spectrograms were displayed in Audacity 3.7.9, and labels were added to the screenshots afterwards. Probe levels were also measured directly from the WAV files with a short Python analysis, as the RMS level within the probe's frequency band during each probe.

Both clips used a 1000 Hz masker with a digital amplitude of 0.5. The probe amplitude was 0.06 in Clip 1 (18.4 dB below the masker) and 0.03 in Clip 2 (24.4 dB below the masker), with the same probe level used across conditions within each clip. These are digital level differences, not calibrated sound-pressure levels.

Audio Clip 1 — `3036837980_Audio1_temporal.wav` (6.2 s, temporal masking). The masker was 1000 Hz and lasted 100 ms; the probe was 1000 Hz and lasted 40 ms. In the table, the time range identifies the masker and the time after the arrow identifies probe onset.

Time (s)	Event	Condition
0.50–0.60 → 0.62	masker → probe	short gap, **20 ms**
1.50–1.60 → 1.80	masker → probe	long gap, **200 ms**
2.50	probe only	control (probe alone)
3.50–3.60 → 3.80	masker → probe	long gap, 200 ms (repeat)
4.50–4.60 → 4.62	masker → probe	short gap, 20 ms (repeat)
5.50	probe only	control (repeat)

Audio Clip 2 — `3036837980_Audio2_frequency.wav` (7.0 s, frequency masking). The masker was 1000 Hz and lasted 500 ms. Each probe lasted 300 ms and, in the masked conditions, started 100 ms after masker onset, so it occurred entirely within the masker. For overlapping events, the first time range identifies the masker and the second identifies the probe; single times identify probe onset in the controls.

Time (s)	Event	Condition
0.40–0.90 with 0.50–0.80	masker + probe **1200 Hz**	near frequency
1.30	probe 1200 Hz only	control
2.00–2.50 with 2.10–2.40	masker + probe **4000 Hz**	far frequency
2.90	probe 4000 Hz only	control
3.60–4.10 with 3.70–4.00	masker + probe 4000 Hz	far (repeat)
4.50	probe 4000 Hz only	control (repeat)
5.20–5.70 with 5.30–5.60	masker + probe 1200 Hz	near (repeat)
6.10	probe 1200 Hz only	control (repeat)

For the listening test, the listener used headphones at a fixed volume in a quiet room. Short excerpts were presented in random order without revealing the condition. After each excerpt, the listener reported whether the probe was **clearly heard**, **faint or uncertain**, or **not heard**. Each condition was presented six times. Masker-only catch trials were included to check for responses when no probe was present.

1. Spectrum analysis

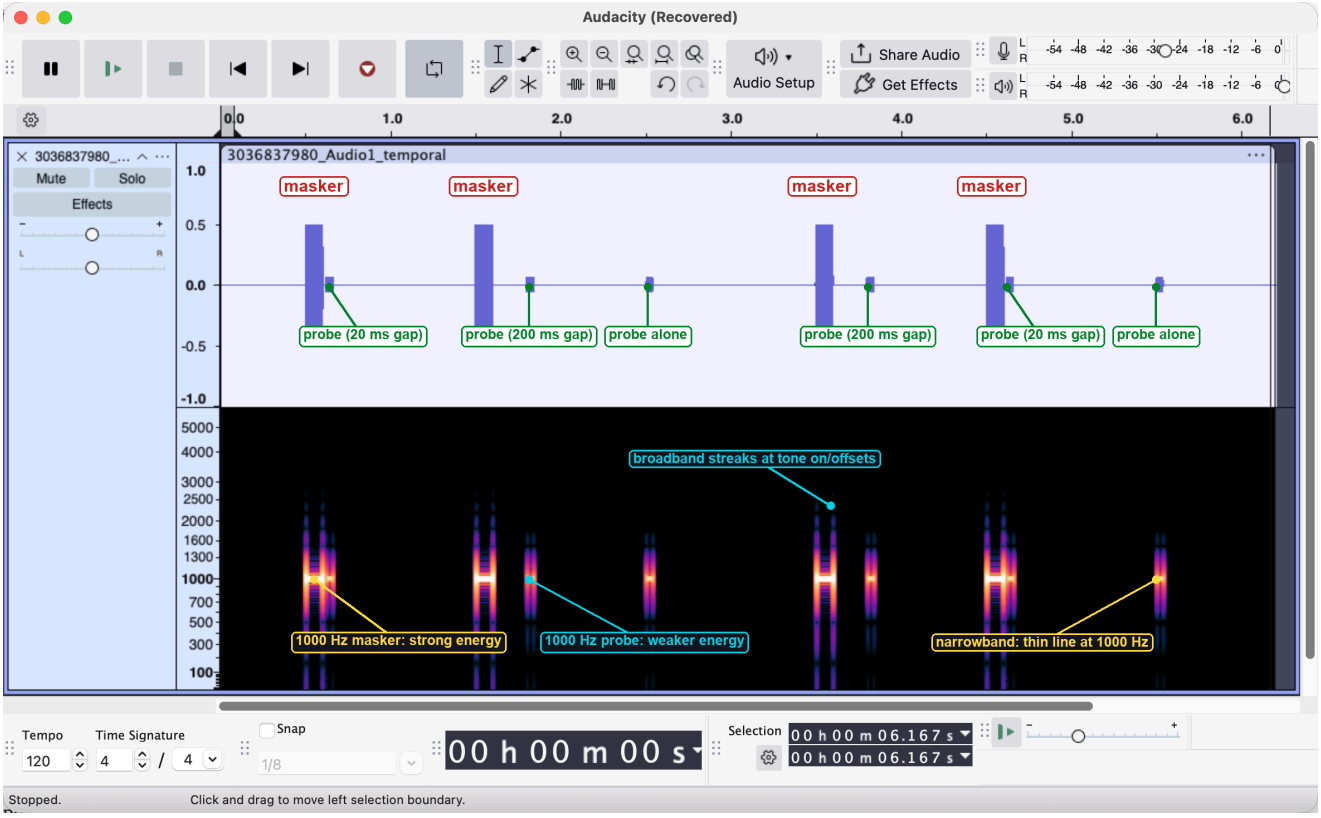


Figure 1. Audio Clip 1: annotated waveform and spectrogram for the temporal masking test.

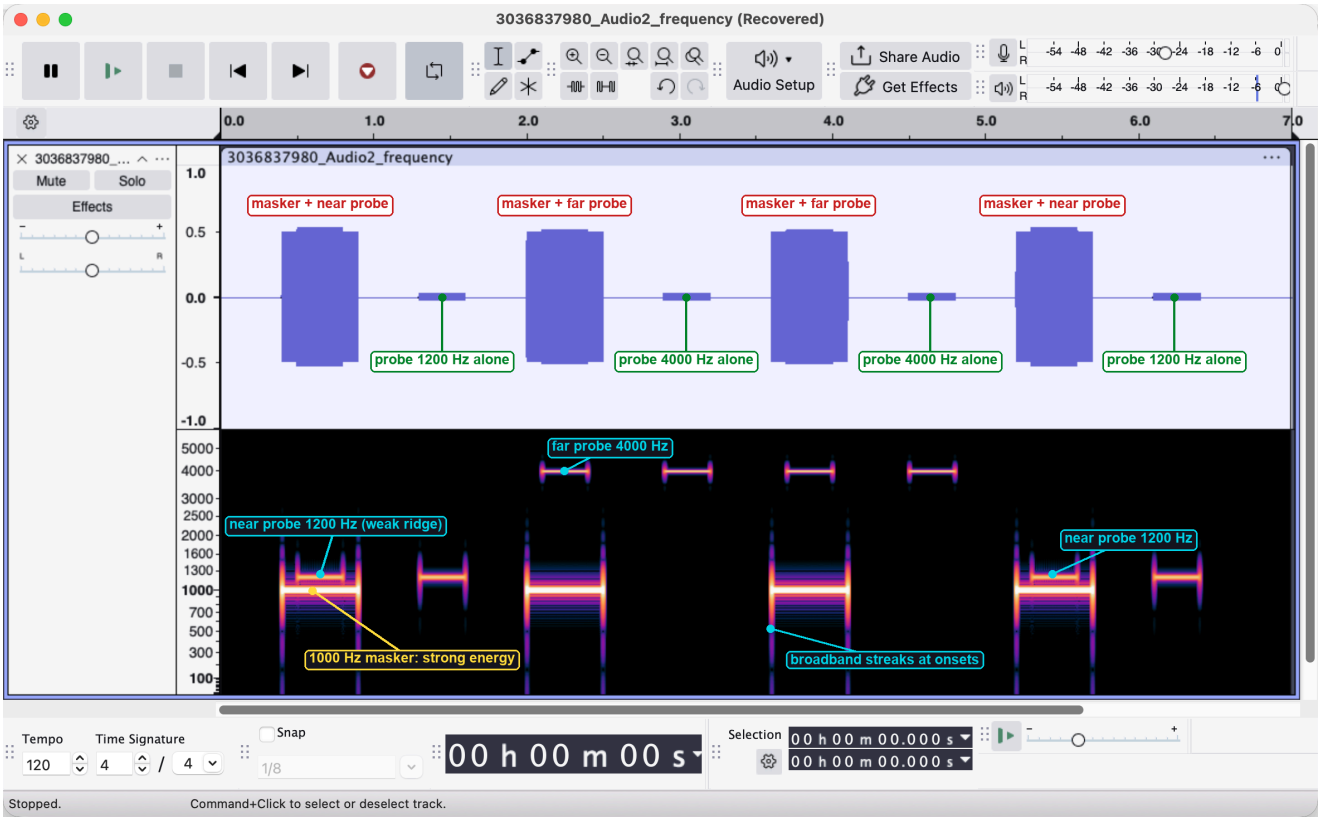


Figure 2. Audio Clip 2: annotated waveform and spectrogram for the frequency masking test.

In each figure, the upper waveform shows digital amplitude over time. The tall blocks correspond to the loud maskers, while the smaller signals show the quiet probes where they occur alone or after a masker. The waveform shows when a sound occurs and its amplitude, but it does not directly identify its frequency. In the lower spectrogram, time runs horizontally and frequency vertically; brighter colours indicate stronger energy at a given time and frequency.

In Figure 1, the bright 1000 Hz regions are the strong maskers, and the weaker patches at the same frequency are the shorter probes. In Figure 2, the strong horizontal ridge at 1000 Hz is the masker; weaker ridges at 1200 Hz and 4000 Hz correspond to the probes. The mostly horizontal, narrowband ridges are characteristic of sine tones. Short vertical streaks at tone boundaries spread across a wider range of frequencies, reflecting the limited duration and onset or offset of the tones rather than a separate continuous broadband sound.

2. Temporal masking — Audio Clip 1

The gap from masker offset to probe onset was **20 ms** in the short-gap condition and **200 ms** in the long-gap condition. The same probe level was used in both conditions and in the probe-alone control. The following are the results of the second temporal test, with six presentations per condition.

Condition	Clearly heard	Faint or uncertain	Not heard
Probe alone	6	0	0
After 200 ms gap	6	0	0
After 20 ms gap	4	2	0
Catch: masker only	0	0	6

The probe was clearly heard in all six presentations when played alone and after the 200 ms gap. After the 20 ms gap, it was clearly heard four times and faint or uncertain twice. This shows reduced clarity at the shorter gap, although the probe was not completely inaudible. No probe was reported in the catch trials.

Figure 1 shows a separate weak 1000 Hz probe after the masker in both gap conditions. The measured probe-band level (see Method) is approximately −28.0 dBFS in each condition, so the change in the listener's response cannot be explained by a lower probe level in the short-gap condition.

This pattern is consistent with **forward temporal masking**: after a strong sound ends, the threshold for detecting a following weak sound remains elevated for a short time and then decreases. The 20 ms probe arrives while the masking effect is stronger; by 200 ms, it has weakened enough for this probe to be clearly heard. In perceptual audio coding, an encoder can use a temporal masking model to allow coarser quantisation, and therefore potentially fewer bits, where the resulting noise remains below the masking threshold.

3. Frequency masking — Audio Clip 2

The 1200 Hz and 4000 Hz probes had the same digital amplitude and each was tested with the 1000 Hz masker and alone. Each condition was presented six times.

Condition	Clearly heard	Faint or uncertain	Not heard
Masker + nearby probe (1200 Hz)	0	0	6
Masker + distant probe (4000 Hz)	6	0	0
1200 Hz probe alone	6	0	0
4000 Hz probe alone	6	0	0
Catch: masker only	0	0	6

The nearby probe was not heard in any of the six presentations with the masker, whereas the distant probe was clearly heard in all six. Both probes were clearly heard when played alone. Figure 2 nevertheless shows the weak 1200 Hz ridge during the masker, around 0.5 s and 5.3 s, confirming that the nearby probe is physically present even when it is not heard.

This is consistent with **simultaneous frequency masking**: a strong sound raises the threshold for detecting a weaker sound occurring at the same time, especially at nearby frequencies. The auditory system analyses sound through overlapping frequency bands, often described as critical bands. The 1200 Hz probe produces substantial excitation in regions responding to the 1000 Hz masker, so it is harder to distinguish; the 4000 Hz probe excites a more distant region and remains audible in this test.

A perceptual audio encoder uses a psychoacoustic model to estimate masking thresholds across frequency bands. It can allocate fewer bits to masked components or use coarser quantisation where the resulting noise stays below the estimated threshold, while preserving more detail where it is audible.

These results describe one listener under the test conditions; they do not establish a general hearing threshold for other listeners or playback levels.